

1) Explain feasibility analysis and the four tests for feasibility analysis with example:

Feasibility Analysis

Feasibility analysis checks if a project is practical, useful, and worth doing before starting development.

The 4 Main Types of Feasibility:

1. Operational Feasibility

Checks if the system will work well in real life and if people will accept and use it.

● Example:

A new software is installed in an office, but if employees find it confusing, it's not operationally feasible.

2. Technical Feasibility

Checks if we have the technology, tools, and skilled people to build the system.

● Example:

A team wants to build an AI tool, but no one knows how to work with AI – not technically feasible.

3. Schedule Feasibility

Checks if the project can be completed on time.

Example:

If a project usually takes 2 months, but the plan is to finish it in 2 weeks – it's not schedule feasible.

4. Economic Feasibility

Checks if the project is cost-effective – whether it saves or earns more money than it costs.

● Example:

If building a system costs \$1,000 but saves \$600 every year, then it pays off in less than 2 years – it's economically feasible.

Suppose PSTU wants to impose smartcard to every student for identity and bill/fees paying issues. Discuss the feasibility of this smartcard project as if you are a member of PSTU authority.

✓ Smartcard Project Feasibility for PSTU (Short)

PSTU plans to introduce smartcards for student identity and fee payments. After analysis:

- **Operational Feasibility:** Students and staff will accept it as it simplifies ID and payment tasks.
- **Technical Feasibility:** PSTU has an IT team; with some support, the system can be implemented.
- **Schedule Feasibility:** The project can be completed within 6–8 months before the next semester.
- **Economic Feasibility:** Though initial costs exist, it will reduce manual work and save money long-term.

✓ **Conclusion:** The smartcard project is feasible and should be implemented.

✓ Cost-Benefit Analysis

▼ Costs

- **Development Costs:** One-time (e.g., software setup)
- **Operating Costs:** Recurring
 - **Fixed:** Regular (e.g., salary)
 - **Variable:** Based on use (e.g., electricity)

▲ Benefits

- **Tangible:** Measurable (e.g., time/money saved)
- **Intangible:** Not measurable (e.g., satisfaction)

Here's a **brief explanation** of each technique:

✓ 1. Net Benefit Analysis

Compares **total benefits** and **total costs** to calculate the **net gain**.

● *Formula:* Net Benefit = Total Benefits – Total Costs

✓ If the result is positive, the project is beneficial.

✓ 2. Payback Analysis

Measures how **long** it takes to recover the initial investment.

● *Example:* If a project costs \$10,000 and saves \$2,000/year → Payback = 5 years

✓ Shorter payback period = better.

✓ 3. Return on Investment (ROI)

Shows **profit percentage** earned from the investment.

● *Formula:* $\text{ROI} = (\text{Net Benefit} / \text{Total Cost}) \times 100\%$

✓ Higher ROI = more profitable project.

✓ 4. Cash Flow Analysis

Tracks **inflow and outflow of money** over time.

● Shows when you earn or spend money during the project.

✓ Helps manage budget and financial planning.